

Math 122: Exam 1 - Practice

Instructions:

- Show all work: Full credit requires complete, correct steps and proper math notation.
 - Points: Each question is worth 10 points.
 - Formula Sheet: Permitted (formulas only; no worked examples).
 - Calculators: Basic, non-graphing calculators allowed.
-

Practice 1

Problem 1 (Section 5.4)

Find the exact arc length of the curve

$$y = 3x^{3/2} - 1$$

over the interval from $x = 0$ to $x = 1$.

Problem 2 (Section 6.5)

Find the limit:

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{\sin x}$$

Problem 3 (Section 6.5)

Find the limit:

$$\lim_{x \rightarrow +\infty} x e^{-x}$$

Problem 4 (Section 6.5)

Find the limit:

$$\lim_{x \rightarrow 0} (e^x + x)^{1/x}$$

Problem 5 (Section 6.7)

Find $\frac{dy}{dx}$:

$$y = \sin^{-1}(1/x)$$

Problem 6 (Section 6.7)

Find $\frac{dy}{dx}$:

$$y = e^x \sec^{-1} x$$

Problem 7 (Section 6.7)

Evaluate the integral:

$$\int \frac{dx}{\sqrt{1-4x^2}}$$

Problem 8 (Section 6.7)

Evaluate the integral:

$$\int \frac{\sec^2 x \, dx}{\sqrt{1 - \tan^2 x}}$$

Problem 9 (Section 6.7)

Evaluate the definite integral:

$$\int_1^{\sqrt{3}} \frac{\sqrt{\tan^{-1} x}}{1 + x^2} \, dx$$

Problem 10 (Section 6.7)

Evaluate the integral:

$$\int \frac{\sin \theta}{\cos^2 \theta + 1} \, d\theta$$

Practice 2**Problem 1**

Find the exact arc length of the curve over the interval from $x = 0$ to $x = 1$:

$$y = (2x + 1)^{3/2}$$

Problem 2

Find the limit:

$$\lim_{x \rightarrow 0} \frac{\tan(2x)}{\ln(1+x)}$$

Problem 3

Find the limit:

$$\lim_{x \rightarrow +\infty} x^2 e^{-2x}$$

Problem 4

Find the limit:

$$\lim_{x \rightarrow 0^+} (1+2x)^{1/x}$$

Problem 5

Find $\frac{dy}{dx}$:

$$y = \cos^{-1}\left(\frac{1}{x^2}\right)$$

Problem 6

Find $\frac{dy}{dx}$:

$$y = \sin^{-1} x \cdot \cos^{-1}(x)$$

Problem 7

Evaluate the integral:

$$\int \frac{dx}{1 + 9x^2}$$

Problem 8

Evaluate the integral:

$$\int \frac{x \, dx}{\sqrt{1 - x^4}}$$

Problem 9

Evaluate the definite integral:

$$\int_0^1 \frac{(\sin^{-1} x)^3}{\sqrt{1 - x^2}} \, dx$$

Problem 10

Evaluate the integral:

$$\int \frac{e^x}{1 + e^{2x}} dx$$